Mobile communication process

User equipment (UE)
- Mobile Device | user
- SIM/USIM for Subscriber Identity
- Radio Interface: Uu [Air Interface]

Role: endpoint for user communication

initiates attach via

Manages handover to maintain

Access Network (RAN) (gNodeB | eNodeB)
- Radio Access network (RAN)
- Manages Radio Resources & Scheduling
- Connects UE to the core network

CRITICAL STAGE 1:
RAN Radio Link & Scheduling
- Radio Link monitoring & scheduling
- Link adaptation & power control
- Interference management & throughput control

routes data through

Service Delivery (IMS Internet, QoS)
- IMS: Voice | VoLTE | messaging
- Internet & external Data networks [PDN]
- QoS: ensures Quality of service delivery

ensures QoS for

Mobility management
- Tracks UE location & Registration
- manages Handover & mobility procedures
- maintains session continuity on move

CRITICAL STAGE 2:
Mobility management / Handover
- Handover decision & execution for seamless mobility
- Load balancing & cell reselection
- minimizes call drop | connectivity loss

authenticate through

coordinates mobility context with core

Core Network (AMF, SMF, UPF, UDM | HSS)
- AMF: Access & mobility management Function
- SMF: Session management Function
- UPF: User plane Function
- UDM | HSS: Subscriber Data & Authentication

CRITICAL STAGE 3:
Core QoS Enforcement (UPF | SMF)
- QoS Policy enforcement & traffic shaping
- Bearer | Session management & packet routing
- Ensures data Priority & Service quality

ensures QoS for